

SYSC 5108 Exam Study

Assignment 3, Question 4

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Question Summary

The network has one sensing neuron, three hidden processing neurons, and one processing output neuron:

$$1 \rightarrow 2 \rightarrow \{3, 4\} \rightarrow 5.$$

All processing neurons, including the output neuron, use ReLU activation. The input is

$$x_1 = 0.5$$

and the desired output is

$$t = 0.9.$$

The question asks for the forward output, the backpropagation δ values, the gradients with respect to all weights, updated weights for learning rate $\alpha = 0.1$, and the reduction in error after the update.

Course and Textbook Cross-References

- **Chapter 4 slides, Forward Propagation Generalization:** each layer computes a linear/affine weighted sum and then applies an activation.
- **Chapter 4 slides, Hidden Units:** ReLU is

$$\phi(z) = \max(0, z).$$

- **Chapter 4 slides, Chain Rule and Backward Propagation:** backpropagation recursively computes gradients using the chain rule.
- **Goodfellow, Bengio, and Courville, *Deep Learning*, Chapter 6:** feedforward networks compute affine transformations followed by activation functions; ReLU units use $g(z) = \max(0, z)$, and gradients are propagated backward through the computation graph.

Given Network Parameters

From the diagram:

$$\begin{aligned}w_{2,1} &= 0.3, & w_{2,0} &= 0.1, \\w_{3,2} &= 0.2, & w_{3,0} &= 0.1, \\w_{4,2} &= 0.5, & w_{4,0} &= 0.1, \\w_{5,3} &= 0.4, & w_{5,4} &= 0.6, & w_{5,0} &= 0.1.\end{aligned}$$

Use squared-error loss with the usual $\frac{1}{2}$ factor:

$$E = \frac{1}{2}(t - \hat{y})^2.$$

The ReLU derivative is

$$\phi'(z) = \begin{cases} 1, & z > 0, \\ 0, & z < 0. \end{cases}$$

All weighted sums in this problem are positive, so every ReLU derivative used in backpropagation is 1.

Part (a): Forward Pass

Neuron 2

$$\begin{aligned} z_2 &= w_{2,0} + w_{2,1}x_1 \\ &= 0.1 + (0.3)(0.5) \\ &= 0.25. \end{aligned}$$

Because $z_2 > 0$,

$$a_2 = \phi(z_2) = 0.25.$$

Neuron 3

$$\begin{aligned} z_3 &= w_{3,0} + w_{3,2}a_2 \\ &= 0.1 + (0.2)(0.25) \\ &= 0.15, \end{aligned}$$

so

$$a_3 = \phi(z_3) = 0.15.$$

Neuron 4

$$\begin{aligned} z_4 &= w_{4,0} + w_{4,2}a_2 \\ &= 0.1 + (0.5)(0.25) \\ &= 0.225, \end{aligned}$$

so

$$a_4 = \phi(z_4) = 0.225.$$

Neuron 5

$$\begin{aligned} z_5 &= w_{5,0} + w_{5,3}a_3 + w_{5,4}a_4 \\ &= 0.1 + (0.4)(0.15) + (0.6)(0.225) \\ &= 0.1 + 0.06 + 0.135 \\ &= 0.295. \end{aligned}$$

Therefore,

$$\hat{y} = a_5 = \phi(z_5) = 0.295.$$

$$\boxed{\hat{y} = 0.295}$$

Part (b): Delta Values

Use the convention

$$\delta_j = \frac{\partial E}{\partial z_j}.$$

Output neuron

For the output neuron,

$$\begin{aligned}\delta_5 &= \frac{\partial E}{\partial z_5} \\ &= (\hat{y} - t)\phi'(z_5) \\ &= (0.295 - 0.9)(1) \\ &= -0.605.\end{aligned}$$

Hidden neurons

For a hidden neuron j ,

$$\delta_j = \phi'(z_j) \sum_k w_{k,j} \delta_k,$$

where the sum is over neurons k that receive input from neuron j .

Neuron 4:

$$\delta_4 = \phi'(z_4)w_{5,4}\delta_5 = (1)(0.6)(-0.605) = -0.363.$$

Neuron 3:

$$\delta_3 = \phi'(z_3)w_{5,3}\delta_5 = (1)(0.4)(-0.605) = -0.242.$$

Neuron 2:

$$\begin{aligned}\delta_2 &= \phi'(z_2) (w_{3,2}\delta_3 + w_{4,2}\delta_4) \\ &= (1) [(0.2)(-0.242) + (0.5)(-0.363)] \\ &= -0.0484 - 0.1815 \\ &= -0.2299.\end{aligned}$$

$\delta_5 = -0.605, \quad \delta_4 = -0.363, \quad \delta_3 = -0.242, \quad \delta_2 = -0.2299$

Part (c): Gradients

For a weight $w_{j,i}$ feeding activation a_i into neuron j ,

$$\frac{\partial E}{\partial w_{j,i}} = \delta_j a_i.$$

For a bias weight $w_{j,0}$, the input is 1, so

$$\frac{\partial E}{\partial w_{j,0}} = \delta_j.$$

Weight	Gradient	Calculation
$w_{5,3}$	-0.09075	$(-0.605)(0.15)$
$w_{5,4}$	-0.136125	$(-0.605)(0.225)$
$w_{5,0}$	-0.605	δ_5
$w_{4,2}$	-0.09075	$(-0.363)(0.25)$
$w_{4,0}$	-0.363	δ_4
$w_{3,2}$	-0.0605	$(-0.242)(0.25)$
$w_{3,0}$	-0.242	δ_3
$w_{2,1}$	-0.11495	$(-0.2299)(0.5)$
$w_{2,0}$	-0.2299	δ_2

Part (d): Updated Weights

Use gradient descent:

$$w_{\text{new}} = w_{\text{old}} - \alpha \frac{\partial E}{\partial w}, \quad \alpha = 0.1.$$

Weight	Old value	Gradient	New value
$w_{2,1}$	0.300000	-0.114950	0.311495
$w_{2,0}$	0.100000	-0.229900	0.122990
$w_{3,2}$	0.200000	-0.060500	0.206050
$w_{3,0}$	0.100000	-0.242000	0.124200
$w_{4,2}$	0.500000	-0.090750	0.509075
$w_{4,0}$	0.100000	-0.363000	0.136300
$w_{5,3}$	0.400000	-0.090750	0.409075
$w_{5,4}$	0.600000	-0.136125	0.613613
$w_{5,0}$	0.100000	-0.605000	0.160500

Part (e): Error Reduction

The original error is

$$\begin{aligned}
E_{\text{old}} &= \frac{1}{2}(t - \hat{y})^2 \\
&= \frac{1}{2}(0.9 - 0.295)^2 \\
&= \frac{1}{2}(0.605)^2 \\
&= 0.1830125.
\end{aligned}$$

Now do a new forward pass using the updated weights.

Neuron 2:

$$z_2^{\text{new}} = 0.12299 + (0.311495)(0.5) = 0.2787375, \quad a_2^{\text{new}} = 0.2787375.$$

Neuron 3:

$$z_3^{\text{new}} = 0.1242 + (0.20605)(0.2787375) \approx 0.181634, \quad a_3^{\text{new}} \approx 0.181634.$$

Neuron 4:

$$z_4^{\text{new}} = 0.1363 + (0.509075)(0.2787375) \approx 0.278198, \quad a_4^{\text{new}} \approx 0.278198.$$

Neuron 5:

$$z_5^{\text{new}} = 0.1605 + (0.409075)(0.181634) + (0.6136125)(0.278198) \\ \approx 0.405508.$$

Thus

$$\hat{y}_{\text{new}} \approx 0.405508.$$

The new error is

$$E_{\text{new}} = \frac{1}{2}(0.9 - 0.405508)^2 \\ \approx 0.122261.$$

Therefore, the reduction in error is

$$E_{\text{old}} - E_{\text{new}} = 0.1830125 - 0.122261 \\ \approx 0.060751.$$

Error reduction ≈ 0.06075

Final Answer

$\hat{y} = 0.295$

$\delta_5 = -0.605, \quad \delta_4 = -0.363, \quad \delta_3 = -0.242, \quad \delta_2 = -0.2299$

Weight	Updated value
$w_{2,1}$	0.311495
$w_{2,0}$	0.122990
$w_{3,2}$	0.206050
$w_{3,0}$	0.124200
$w_{4,2}$	0.509075
$w_{4,0}$	0.136300
$w_{5,3}$	0.409075
$w_{5,4}$	0.613613
$w_{5,0}$	0.160500

$E_{\text{old}} = 0.1830125, \quad E_{\text{new}} \approx 0.122261, \quad E_{\text{old}} - E_{\text{new}} \approx 0.06075$

Exam Notes

- Since every $z_j > 0$, every ReLU derivative in this problem is 1. That is why the backpropagation is numerically simple.
- The sign convention here is $\delta_j = \partial E / \partial z_j$. With this convention, use $w_{\text{new}} = w - \alpha \partial E / \partial w$.
- The output 0.295 is below the target 0.9, so the gradients are negative and gradient descent increases the relevant weights.